MvLogistics SDK (C or C++)

Developer Guide

Legal Information

Contents

Chapter 1 Overview

The MvLogisticsSDK is a software development kit used to encapsulate machine vision camera module, code reader module, panoramic camera module, line scan camera module, 3D camera module, weighing module, and fusion module, for meeting the requirements of different scenarios in logistics industry.

1.1 Introduction

This manual mainly introduces the MvLogisticsSDK based on C/C++ language, which provides several APIs, MFC demo, and WinForm demo to configure the parameters of different module combinations for transmitting package information.

1.2 Development Environment

The development environment of MvLogisticsSDK is shown in the table below.

Item	Required
Operating System	Microsoft® Windows XP (32-bit)/Windows 7 (32/64-bit)/Windows 10 (32/64-bit)
Processor	Intel Pentium IV 3.0GHz CPU and above
Random Access Memory (RAM)	4GB and above

1.3 Update History

This page lists the summary of changes, including content adding, editing, and deleting, during SDK version updating process.

Summary of Changes in Version 1.1.0_03/2020

Version	Summary of Changes
Version 1.1.0_03/2020	1. Added API of getting NoRead images of code reader in callback function: ***MV_LGS_RegisterNoReadImageCB*** .
	2. Added API of getting camera detailed information in XML configuration file: ***MV_LGS_GetXmlCfgCamInfo*** .
	3. Extended the barcode information structure ***MVLGS_CODE_INFO*** :

Version	Summary of Changes
	added one member **strSerialNumber** (camera serial No.).
	4. Extended the package volume information structure ***MVLGS_VOLUME_INFO*** : added two members **nObjEnterSysTime** (volume measurement start timestamp) and **nObjLeaveSysTime** (volume measurement end timestamp).
	5. Extended the package information structure ***MVLGS_PACKAGE_INFO*** : added three members: **bIsFirstCodeOut** (whether to set the barcode as first output), **llTriggerStartTimeStamp** (camera trigger start timestamp), and **llTriggerEndTimeStamp** (camera trigger end timestamp).

Summary of Changes in Version 1.0.0_03/2019

New document.

1.4 Notice

- To configure the parameters of machine vision camera, panoramic camera, and line scan camera, you should install and run the Machine Vision Software (MVS) first, then connect to the camera to check if it can acquire image, and configure the parameters. You can save the configured parameters to the camera, so each time when the camera is powered on, these saved parameters will take effect.
- To configure the parameters of code reader, you should install and run the ID Machine Vision Software (IDMVS) first, then connect to the code reader to check if it can acquire image with code information, and configure the parameters. You can save the configured parameters to the camera, so each time when the camera is powered on, these saved parameters will take effect.
- Before operating the 3D camera, you should use corresponding demo to calibrate and save the camera parameters.
- If the Machine Vision Software (MVS) version is V2.3.0 or below, you should add the installation directory to environment variables.
- To install the SoftDog driver, you should insert the iMVS1001 or iMVS-6000 SoftDog.
- For the operating system of Microsoft® Windows XP, you should install the Microsoft .NET Framework 3.5 Service Pack 1.

Chapter 2 Transmit Package Information

In the logistics industry, a fast acquisition and transmission process of package information, such as bar code information on waybill, is quite important to improve the logistics management efficiency. Here you can follow the API call process to combine and configure multiple camera or algorithm modules to get and transmit package information.

Before You Start

Make sure the Machine Vision Software (MVS) and ID Machine Vision Software (IDMVS) have installed.

Steps

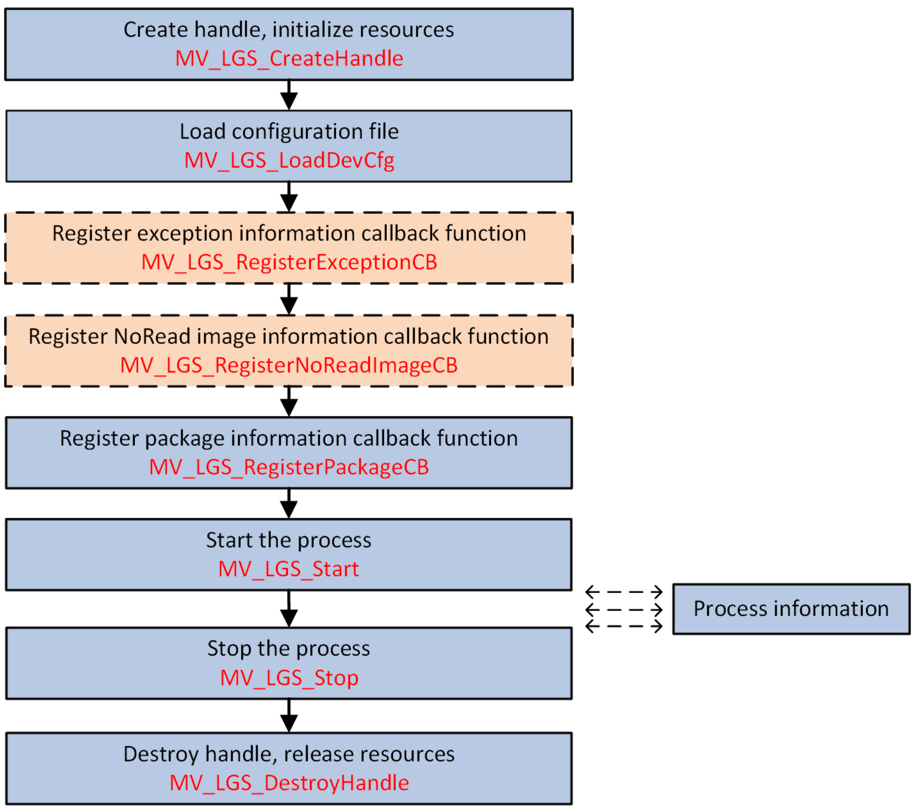

Figure 2-1 Programming Flow of Transmitting Package Information

1. Call ***MV_LGS_CreateHandle*** to create the handle and initialize resources.
2. Call ***MV_LGS_LoadDevCfg*** to load the device configuration file.

Note

The configuration files, including basic parameters and fusion module parameters, should be configured in the XML messages of ***XML_MvLogisticsSDK*** and ***XML_FusionCfg*** , as the APIs used for setting and saving parameters are not provided.

3. **Optional:** Call ***MV_LGS_RegisterExceptionCB*** to register the exception information callback function.

   The exception information is returned in the structure ***MVLGS_EXCEPTION_INFO*** of callback function **cbException**.
4. **Optional:** Call ***MV_LGS_RegisterNoReadImageCB*** to register the NoRead image information callback function.

   The NoRead image information is returned in the structure ***MVLGS_IMAGE_OUTPUT_INFO*** of callback function **cbNoReadImageOutput**.
5. Call ***MV_LGS_RegisterPackageCB*** to register the package information callback function.

   The package information is returned in the structure ***MVLGS_PACKAGE_INFO*** of callback function **cbOutput**.
6. Call ***MV_LGS_Start*** to start the process.
7. Process the received package information.
8. Call ***MV_LGS_Stop*** to stop the process.
9. Call ***MV_LGS_DestroyHandle*** to destroy the handle and release resources.

Chapter 3 API Reference

3.1 MV_LGS_CreateHandle

Create the handle, and initialize resources.

API Definition

```
int MV_LGS_CreateHandle(
  void          **handle
)
```

Parameters

handle

[IN] [OUT] Handle

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

3.2 MV_LGS_DestroyHandle

Destroy the handle, and release resources.

API Definition

```
int MV_LGS_DestroyHandle(
  void            *handle
)
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

See Also

MV_LGS_CreateHandle

3.3 MV_LGS_GetVersion

Get the SDK version No.

API Definition

```
int MV_LGS_GetVersion(
)
```

Return Value

Return SDK version No. on success, for example, 0×01010000 indicates V1.1.0.0, and the format is as follows:

Main	Sub	Revision	Test
8bits	8bits	8bits	8bits

3.4 MV_LGS_GetXmlCfgCamInfo

Get the camera detailed information in XML configuration file.

API Definition

```
int MV_LGS_GetXmlCfgCamInfo(
  MVLGS_XML_CFG_CAM_INFO_LIST      *pstXmlCfgCamInfo,
  const char                       *strCfgPath
);
```

Parameters

pstXmlCfgCamInfo

[IN][OUT] List of camera information in configuration file, see structure ***MVLGS_XML_CFG_CAM_INFO_LIST*** for details.

strCfgPath

[IN] Configuration file path

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

3.5 MV_LGS_LoadDevCfg

Load device configuration file.

API Definition

```
int MV_LGS_LoadDevCfg(
  void              *handle,
  const char        *strCfgPath
)
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

strCfgPath

[IN] Configuration file path

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

3.6 MV_LGS_RegisterExceptionCB

Get exception information in callback function.

API Definition

```
int MV_LGS_RegisterExceptionCB(
  void          *handle,
  void          *cbException,
  void          *pUser
)
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

cbException

[IN] Callback function for receiving exception information, see the definitions below:

```
void(__stdcall* cbException)(
  MVLGS_EXCEPTION_INFO      *pstEcptInfo,
  void                      *pUser
)
```

pstEcptInfo

[OUT] Exception information, see the structure ***MVLGS_EXCEPTION_INFO*** for details.

pUser

[OUT] Pointer to user

pUser

[IN] Pointer to user

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

3.7 MV_LGS_RegisterPackageCB

Get package information in callback function.

API Definition

```
int MV_LGS_RegisterPackageCB(
  void              *handle,
  void              *cbOutput,
  void              *pUser
)
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

cbOutput

[IN] Callback function for receiving package information, see the definitions below:

```
void(__stdcall* cbOutput)(
  MVLGS_PACKAGE_INFO       *pstPkgInfo,
  void                     *pUser
)
```

pstPkgInfo

[OUT] Package information, see the structure ***MVLGS_PACKAGE_INFO*** for details.

pUser

[OUT] Pointer to user

pUser

[IN] Pointer to user

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

3.8 MV_LGS_RegisterNoReadImageCB

Get NoRead images of code reader in callback function.

API Definition

```
int MV_LGS_RegisterNoReadImageCB(
  void              *handle,
  void              *cbNoReadImageOutput,
  void              *pUser
)
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

cbNoReadImageOutput

[IN] Callback function for receiving NoRead images, see the definitions below:

```
void(__stdcall* cbNoReadImageOutput)(
  MVLGS_IMAGE_OUTPUT_INFO    *pstImageOutPutInfo,
  void                       *pUser
)
```

pstImageOutPutInfo

[IN] NoRead image information, see the structure ***MVLGS_IMAGE_OUTPUT_INFO*** for details.

pUser

[IN] Pointer to user

pUser

[IN] Pointer to user

Return Value

Return *MV_LGS_OK* for success, and return ***Error Code*** for failure.

Remarks

This API only supports code reader.

3.9 MV_LGS_SetTrigger

Set trigger status. You can call this API to stimulate the trigger under the condition of no actual physical trigger.

API Definition

```
int MV_LGS_SetTrigger(
  void                *handle,
  unsigned int        nTriggerSignal
)
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

nTriggerSignal

[IN] Trigger status: 1-trigger started, 0-trigger ended

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

3.10 MV_LGS_Start

Start the process.

API Definition

```
int MV_LGS_Start(
  void        *handle
)
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

3.11 MV_LGS_Stop

Stop the process.

API Definition

```
int MV_LGS_Stop(
  void        *handle
```

Parameters

handle

[IN] Handle, which is returned by ***MV_LGS_CreateHandle*** .

Return Value

Return *MV_LGS_OK* on success, and return ***Error Code*** on failure.

Appendix A. Appendixes

A.1 Data Structure

A.1.1 MVLGS_CODE_INFO

Barcode information structure.

Structure Definition

```
typedef struct{
  unsigned char         strCode[MVLGS_MAX_CODECHARATERLEN/*4096*/];
  int                   nLen;
  MVLGS_CODE_TYPE       enBarType;
  MVLGS_POINT_I         stCornerPt[4];
  int                   nAngle;
  int                   nFilterFlag;
  char                  strSerialNumber[32];
  unsigned int          nReserved[23];
}MVLGS_CODE_INFO;
```

Members

strCode

Code character

nLen

Character length

enBarType

Barcode type, see the enumeration definition ***MVLGS_CODE_TYPE*** for details.

stCornerPt

Barcode position, see the structure ***MVLGS_POINT_I*** for details.

nAngle

Barcode angle, ranges from 0 to 3600

nFilterFlag

Filtered barcode mark: 0-normal barcode, 1-filtered barcode

strSerialNumber

Camera serial No.

nReserved

Reserved.

A.1.2 MVLGS_CODE_LIST

Barcode information list structure.

Structure Definition

```
typedef struct{
  int                  nCodeNum;
  MVLGS_CODE_INFO      stCodeInfo[MVLGS_MAX_CODENUM];
  MVLGS_IMAGE_INFO     stImage;
  unsigned char        *pImageWaybill;
  unsigned int         nImageWaybillLen;
  unsigned int         nReserved[8];
}MVLGS_CODE_LIST;
```

Members

nCodeNum

The number of barcodes.

stCodeInfo

Barcode information, see the structure ***MVLGS_CODE_INFO*** for details.

stImage

Original image information, see the structure ***MVLGS_IMAGE_INFO*** for details.

pImageWaybill

Image matting buffer, which is allocated by SDK

nImageWaybillLen

The image size

nReserved

Reserved.

A.1.3 MVLGS_EXCEPTION_INFO

Exception information structure.

Structure Definition

```
typedef struct{
  unsigned int    nExceptionID;
  char            strExceptionDes[256];
  unsigned int    nReserved[16];
}MVLGS_EXCEPTION_INFO;
```

Members

nExceptionID

Exception ID

strExceptionDes

Exception description

nReserved

Reserved.

A.1.4 MVLGS_IMAGE_INFO

Output image information structure.

Structure Definition

```
typedef struct{
  unsigned char       *pImageBuf;
  unsigned int        nImageLen;
  MVLGS_IMAGE_TYPE    enImageType;
  unsigned short      nWidth;
  unsigned short      nHeight;
  unsigned int        nTriggerIndex;
  unsigned int        nFrameNum;
  unsigned int        nDevTimeStampHigh;
  unsigned int        nDevTimeStampLow;
  unsigned int        nReserved[8];
}MVLGS_IMAGE_INFO;
```

Members

pImageBuf

Input image buffer

nImageLen

Original image size

enImageType

Image type, see the enumeration definition ***MVLGS_IMAGE_TYPE*** for details.

nWidth

Image width

nHeight

Image height

nTriggerIndex

Trigger index, it can be used as package No. in special case.

nFrameNum

Frame No.

nDevTimeStampHigh

Timestamp high 32 bits

nDevTimeStampLow

Timestamp low 32 bits

nReserved

Reserved.

A.1.5 MVLGS_IMAGE_OUTPUT_INFO

Camera's output image information structure

Structure Definition

```
typedef struct{
  MVLGS_IMAGE_INFO            stImage;
  char                        strSerialNumber[32];
  unsigned int                nReserved[32];
}MVLGS_IMAGE_OUTPUT_INFO;
```

Members

stImage

Image information, see the structure ***MVLGS_IMAGE_INFO*** for details.

strSerialNumber

Camera serial No.

nReserved

Reserved.

A.1.6 MVLGS_PACKAGE_INFO

Package information structure.

Structure Definition

```
typedef struct{
  bool                bCodeEnable;
  unsigned int        nCodeListNum;
  MVLGS_CODE_LIST     stCodeList[MVLGS_MAX_CODELISTNUM/*24*/];
  bool                bVolumeEnable;
  MVLGS_VOLUME_INFO   stVolumeInfo;
  bool                bWeightEnable;
```

```
  float                  fWeight;
  bool                   bPRMEnable;
  MVLGS_IMAGE_INFO       stPRMImage;
  bool                   bIsFirstCodeOut;
  long long              llTriggerStartTimeStamp;
  long long              llTriggerEndTimeStamp;
  unsigned int           nReserved[27];
}MVLGS_PACKAGE_INFO;
```

Members

bCodeEnable

Nonzero values indicate that the barcode information is included.

nCodeListNum

The number of barcodes lists.

stCodeList

The barcode information list, see the structure ***MVLGS_CODE_LIST*** for details

bVolumeEnable

Nonzero values indicate that the volume information is included.

stVolumeInfo

The volume information, see the structure ***MVLGS_VOLUME_INFO*** for details.

bWeightEnable

Nonzero values indicate that the weight information is included.

fWeight

Weight

bPRMEnable

Nonzero values indicate that the panoramic image is included.

stPRMImage

The panoramic image information, see the structure ***MVLGS_IMAGE_INFO*** for details.

bIsFirstCodeOut

Whether to set the barcode as first output: 1-first output, 0-second output

llTriggerStartTimeStamp

Camera trigger start timestamp

llTriggerEndTimeStamp

Camera trigger end timestamp

nReserved

Reserved.

A.1.7 MVLGS_POINT_I

Position information structure.

Structure Definition

```
typedef struct{
  int         nX;
  int         nY;
}MVLGS_POINT_I;
```

Members

nX

X-coordinate

nY

Y-coordinate

A.1.8 MVLGS_VOLUME_INFO

Package volume information structure

Structure Definition

```
typedef struct{
  unsigned int        nPkgID;
  float               fVolume;
  float               fLength;
  float               fWidth;
  float               fHeight;
  int64_t             nObjEnterSysTime;
  int64_t             nObjLeaveSysTime;
  unsigned int        nReserved[60];
}MVLGS_VOLUME_INFO;
```

Members

nPkgID

Package ID

fVolume

Volume

fLength

Length

fWidth

Width

fHeight

Height

nObjEnterSysTime

Volume measurement start timestamp

nObjLeaveSysTime

Volume measurement end timestamp

nReserved

Reserved.

A.1.9 MVLGS_XML_CFG_CAM_INFO

Structure about camera information in XML configuration file

Structure Definition

```
typedef struct{
  MVLGS_DEVICE_TYPE          enXmlCamType;
  unsigned int               nMacAddrHigh;
  unsigned int               nMacAddrLow;
  unsigned int               nCurrentIp;
  unsigned char              strManufacturerName[32];
  unsigned char              strModelName[32];
  unsigned char              strDeviceVersion[32];
  unsigned char              strSerialNumber[32];
  unsigned char              strUserDefinedName[16];
  bool                       bDeviceOnline;
  bool                       bMainCamera;
  unsigned int               nReserved[16];
}MVLGS_XML_CFG_CAM_INFO;
```

Members

enXmlCamType

Camera type, see details in ***MVLGS_DEVICE_TYPE*** .

nMacAddrHigh

Mac high address

nMacAddrLow

Mac low address

nCurrentIp

Current IP address

strManufacturerName

Manufacturer

strModelName

Model

strDeviceVersion

Device version No.

strSerialNumber

Device serial No.

strUserDefinedName

Custom name

bDeviceOnline

Device status: "true"-online, "false"-offline

bMainCamera

Whether it is the main camera

nReserved

Reserved

A.1.10 MVLGS_XML_CFG_CAM_INFO_LIST

Structure about list of camera information in XML configuration file

Structure Definition

```
typedef struct{
  unsigned int                nXmlCfgCamNum;
  MVLGS_XML_CFG_CAM_INFO      *pstXmlCamInfo[256];
}MVLGS_XML_CFG_CAM_INFO_LIST;
```

Members

nXmlCfgCamNum

Number of cameras in configuration file

pstXmlCamInfo

Camera information, see the structure ***MVLGS_XML_CFG_CAM_INFO***

A.2 Enumeration

A.2.1 MVLGS_CODE_TYPE

Barcode Enumeration Definition

Enumeration Type	Macro Definition Value	Description
MVLGS_CODE_NONE	0	No barcode
MVLGS_CODE_TDCR_DM	1	DM barcode
MVLGS_CODE_TDCR_QR	2	QR barcode
MVLGS_CODE_BCR_EAN8	8	EAN8 barcode
MVLGS_CODE_BCR_UPCE	9	UPCE barcode
MVLGS_CODE_BCR_UPCA	12	UPCA barcode
MVLGS_CODE_BCR_EAN13	13	EAN13 barcode
MVLGS_CODE_BCR_ISBN13	14	ISBN13 barcode
MVLGS_CODE_BCR_CODABAR	20	Codebar
MVLGS_CODE_BCR_ITF25	25	Interleaved 2 of 5 (ITF25)
MVLGS_CODE_BCR_CODE39	39	Code 39
MVLGS_CODE_BCR_CODE93	93	Code 93
MVLGS_CODE_BCR_CODE128	128	Code 128

A.2.2 MVLGS_DEVICE_TYPE

Camera Enumeration Definition

Enumeration Type	Macro Definition Value	Description
MVLGS_IDCAM	0	Industrial camera
MVLGS_CODEREADER	1	Code reader
MVLGS_LINESCAN	2	Line scan camera
MVLGS_PANORAMIC	3	Panoramic camera
MVLGS_VOLUME	4	Volume camera
MVLGS_WEIGHT	5	Weighing module. Now it is not supported

A.2.3 MVLGS_IMAGE_TYPE

Image Enumeration Definition

Enumeration Type	Macro Definition Value	Description
MVLGS_IMAGE_Undefined	0	Undefined type
MVLGS_IMAGE_MONO8	1	Mono8
MVLGS_IMAGE_JPEG	2	JPEG
MVLGS_IMAGE_BMP	3	BMP
MVLGS_IMAGE_RGB24	4	RGB24
MVLGS_IMAGE_BGR24	5	BGR

A.3 XML Message

The XML message contains parameters to be configured in the manual are listed here for reference.

A.3.1 XML_FusionCfg

Fusion module parameters in XML format

```
<?xml version="1.0" encoding="UTF-8"?>
<CtrlConfig>
  <OutputMode help="1: by barcode, 2: end by package, 3: start by package"
hide="0" value="2" type="4" valuelist="1,2,3" index="1" name="output mode"/>
  <MinWaitTime help="output immediately when the number of read barcodes
reached the configured maximum number and the time reached" hide="0" value="0"
type="3" valuelist="1" name="the minimum waiting time"/>
  <MaxWaitTime help="output when the number of read barcodes does not reach the
configured maximum number, while the time reached" hide="0" value="500"
type="3" valuelist="1" name="the maximum waiting time"/>
  <MaxProcWaitTime help="The maximum waiting time for matching, ranges from 0
to 5000ms" hide="0" value="1500" type="3" valuelist="1" name="The maximum
waiting time for weight matching"/>
  <FilterNum help="the number of filtered barcodes, in the condition of
outputted barcodes are valid, it ranges from 0 to 100. The software save the
read barcodes in inside list, if a duplicated barcode is detected, it will be
filtered" hide="0" value="2" type="3" valuelist="1" name="the number of
filtered dumplicated barcedes"/>
  <DelSameCodeEnable help="Enable deleting duplicated barcode" hide="0"
value="1" type="2" name="Enable deleting duplicated barcode"/>
```

```
  <CustomFilterRule>
    <RuleEnable help="enable custom rule" hide="0" value="0" type="2"
name="enable custom rule"/>
    <FilterName help="generate rule name" hide="0" value="default.rule"
type="3" valuelist="3" name="rule name"/>
  </CustomFilterRule>
  <MaxLabelCodeNum help="the maximum number of label barcodes" hide="0"
value="2" type="3" valuelist="1" name="the maximum of barcodes"/>
  <MatchVolumeEnable help="enable volume matching output" hide="0" value="0"
type="2" name="matched volume"/>
  <MatchWeightEnable help="enable weight matching output" hide="0" value="0"
type="2" name="matched weight"/>
</CtrlConfig>
```

A.3.2 XML_MvLogisticsSDK

Message for setting basic parameters of logistics application based on machine vision cameras, and it is in XML format.

```
<?xml version="1.0" encoding="UTF-8"?>
<MvLogisticsRoot>
    <String Name="MvLogisticsSDKVersion">
    <Description>MvLogistics SDK version number</Description>
    <Value>01010000</Value>
  </String>
  <!--Machine vision camera configuration-->
  <Category Name="IndustrialCamera">
    <Integer Name="IDCameraNum">
      <Description>The number of cameras, up to 256 cameras are allowed. Value
range: [0,256], the default value is 0</Description>
      <Value>0</Value>
    </Integer>
    <!--When the inputted number of cameras is larger than 0, the corresponding
camera information is required-->
      <String Name="IDCamera1">
        <Description>Camera 1 serial No.</Description>
        <Value>00800876677</Value>
      </String>
    <!--The number of cameras in one thread rolling-->
    <Integer Name="IDCameraNumInOneThread">
      <Description>The number of cameras in one thread rolling. Value range:
[1,256], the default value is 1.</Description>
      <Value>1</Value>
    </Integer>
  </Category>

  <!--Code reading camera configuration-->
  <Category Name="CodeReadCamera">
    <Integer Name="CRCameraNum">
      <Description>The number of cameras, up to 256 cameras are allowed. Value
```

```
range: [0,256], the default value is 0</Description>
      <Value>1</Value>
    </Integer>
    <!--When the inputted number of cameras is larger than 0, the corresponding
camera information is required-->
    <String Name="CRCamera1">
      <Description>Camera 1 serial No.</Description>
      <Value>217872108</Value>
    </String>
    <!--The number of cameras in one thread rolling-->
    <Integer Name="CRCameraNumInOneThread">
      <Description>The number of cameras in one thread rolling, not support
now. Value range: [1,256], the default value is 1</Description>
      <Value>1</Value>
    </Integer>
  </Category>

  <!--Panoramic camera, up to one camera is supported, if no panoramic camera,
input 0-->
  <Category Name="PanoramicCamera">
    <Integer Name="PRMCameraNum">
      <Description>The number of cameras, value range: [0,1], the default value
is 0</Description>
      <Value>0</Value>
    </Integer>
    <!--When the inputted number of cameras is larger than 0, the corresponding
camera information is required-->
    <String Name="PRMCamera1">
      <Description>Camera 1 serial No.</Description>
      <Value>670094111</Value>
    </String>
  </Category>

  <!--Line scan camera, up to one camera is supported, if no line scan camera,
input 0-->
  <Category Name="LineScanCamera">
    <Integer Name="LSCameraNum">
      <Description>The number of cameras, value range: [0,1], the default value
is 0</Description>
      <Value>0</Value>
    </Integer>
    <!--When the inputted number of cameras is larger than 0, the corresponding
camera information is required-->
    <String Name="LSCamera1">
      <Description>Camera 1 serial No.</Description>
      <Value>670094111</Value>
    </String>
  </Category>

  <!--Volume camera, up to one camera is supported, if no volume camera, input
0-->
  <Category Name="VolumeCamera">
```

```
    <Integer Name="VOLCameraNum">
      <Description>The number of cameras, the default value is 0</Description>
      <Value>0</Value>
    </Integer>
    <!--When the inputted number of cameras is larger than 0, the corresponding
camera information is required-->
    <String Name="VOLCamera1">
      <Description>When the working mode is 13, it is the serial number; for
other working mode, it is Mac address</Description>
      <Value>c42f90ff1992</Value>
    </String>
    <Integer Name="VOLAlgorithmType">
      <Description>The camera working mode, range: 5 and [7,13], the default
value is 5</Description>
      <Value>5</Value>
    </Integer>
  </Category>

  <!--1D barcode parameters configuration (not for code reader)-->
  <Category Name="BarcodeParam">
    <Integer Name="BCR_Enable">
      <Description>Enable barcode or not: 0-disable, 1-enable</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_Ability">
      <Description>Capability set, range: [1,63]; including Code39[1],
Code128[2], CodeBar[4], EAN[8], ITF25[16], CODE93[32]</Description>
      <Value>3</Value>
    </Integer>
    <Integer Name="BCR_PositionXROI">
      <Description>Image ROI X-offset, range: [0,65535], default value: 0</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_PositionYROI">
      <Description>Image ROI Y-offset, range: [0,65535], default value: 0</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_WidthROI">
      <Description>Image ROI width, range: [100,65535], default value: 65535</
Description>
      <Value>65535</Value>
    </Integer>
    <Integer Name="BCR_HeightROI">
      <Description>Image ROI height, range: [40,65535], default value: 65535</
Description>
      <Value>65535</Value>
    </Integer>
    <Integer Name="BCR_MaxWidth">
      <Description>The max. width, range: [0,65535], default value: 3840; this
node is invalid</Description>
```

```
      <Value>3840</Value>
    </Integer>
    <Integer Name="BCR_MaxHeight">
      <Description>The max. height, range: [0,65535], default value: 2748; this
node is invalid</Description>
      <Value>2748</Value>
    </Integer>
    <Integer Name="BCR_LocBarNum">
      <Description>The number of locating barcodes, value range: [1,200], the
default value is 4</Description>
      <Value>4</Value>
    </Integer>
    <Integer Name="BCR_LocWinSize">
      <Description>Locating module window size of barcode region, value range:
[4,65], the default value is 4</Description>
      <Value>4</Value>
    </Integer>
    <Integer Name="BCR_WaitingTime">
      <Description>The max. running time of process, value range: [0,5000], the
default value is 500, unit; ms, when exceeding the limit, it will be forced to
return</Description>
      <Value>500</Value>
    </Integer>
    <Integer Name="BCR_SegQuietW">
      <Description>Quiet zone width, value range: [0,200], the default value is
30</Description>
      <Value>30</Value>
    </Integer>
    <Integer Name="BCR_DfkSizeLowerLimit">
      <Description>The lower limit of filtering fake size (delete the barcode
with width lower than the limit), value range: [0,4000], the default value is
30</Description>
      <Value>30</Value>
    </Integer>
    <Integer Name="BCR_DfkSizeUpperLimit">
      <Description>The upper limit of filtering fake size (delete the barcode
with width larger than the limit), value range: [0,4000], the default value is
2400</Description>
      <Value>2400</Value>
    </Integer>
    <Integer Name="BCR_SaveImageLevel">
      <Description>The sensitivity of saving image, value range: [1,3], the
default value is 1</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="BCR_AppMode">
      <Description>Running mode (Expert/Normal), value range: [1,21477483646],
the default value is 0</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_Distortion">
      <Description>Switch of perspective distortion in expert mode, value
```

```
range: [1,21477483646], the default value is 0</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_WhiteGap">
      <Description>Switch of printing quality in expert mode, value range:
[1,21477483646], the default value is 0</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_Spot">
      <Description>Switch of mirror reflection in expert mode, value range:
[1,21477483646], the default value is 0</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_SampleLevel">
      <Description>Sampling level, value range: [1,8], the default value is 1</
Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="BCR_ImageMorph">
      <Description>Image morph, value range: [0,2], the default value is 0</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="BCR_DelErrFlag">
      <Description>Deleting error flag, value range: [0,1], the default value
is 1</Description>
      <Value>1</Value>
    </Integer>
  </Category>

  <!--2D barcode parameters configuration (not for code reader)-->
  <Category Name="2DCodeParam">
    <Integer Name="TDCR_Enable">
      <Description>Enable or not: 0-disabe, 1-enable</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_Ability">
      <Description>Capability sets</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="TDCR_PositionXROI">
      <Description>Image ROI X-offset, range: [0,65535], default value: 0</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_PositionYROI">
      <Description>Image ROI Y-offset, range: [0,65535], default value: 0</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_WidthROI">
      <Description>Image ROI width, range: [128,65535], default value: 65535</
```

```
Description>
      <Value>65535</Value>
    </Integer>
    <Integer Name="TDCR_HeightROI">
      <Description>Image ROI height, range: [128,65535], default value: 65535</
Description>
      <Value>65535</Value>
    </Integer>
    <Integer Name="TDCR_MaxWidth">
      <Description>The max. width, range: [0,65535], default value: 3840</
Description>
      <Value>3840</Value>
    </Integer>
    <Integer Name="TDCR_MaxHeight">
      <Description>The max. height, range: [0,65535], default value: 2748</
Description>
      <Value>2748</Value>
    </Integer>
    <Integer Name="TDCR_LocCodeNum">
      <Description>The number of RIO outputted by detection module, value
range: [1,1000], the default value is 5.</Description>
      <Value>5</Value>
    </Integer>
    <Integer Name="TDCR_MinBarSize">
      <Description>The min. width and height for blob filtering, value range:
[20,1000], the default value is 40.</Description>
      <Value>40</Value>
    </Integer>
    <Integer Name="TDCR_MaxBarSize">
      <Description>The max. width and height for blob filtering, value range:
[20,1000], the default value is 300.</Description>
      <Value>300</Value>
    </Integer>
    <Integer Name="TDCR_MirrorMode">
      <Description>Enable mirror mode or not: 0-disable, 1-enable, 2-compatible
mode</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_SampleLevel">
      <Description>Decimation ratio, value range: [1,8], the default value is
1</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="TDCR_CodeColor">
      <Description>Barcode color: 0-white background and black bar, 1-black
background and white bar, 2-compatible mode</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_DiscreteFlag">
      <Description>Barcode symbology type: 0-continuous, 1-discrete, 2-
compatible mode</Description>
      <Value>0</Value>
```

```
    </Integer>
    <Integer Name="TDCR_DistortionFlag">
      <Description>QR distortion configuration: 0-disable, 1-enable</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_AdvanceParam">
      <Description>Advanced parameter 1, value range: [0,2147483640], the
default value is 0.</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_WaitingTime">
      <Description>Timeout (ms), when waiting time reaches the timeout, it will
exit. Value range: [0,5000], the default value is 1000.</Description>
      <Value>1000</Value>
    </Integer>
    <Integer Name="TDCR_DebugFlag">
      <Description>Enable bug information or not: 0-disable, 1-enable</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_AdvanceParam2">
      <Description>Advanced parameter 2, value range: [0,2147483640], the
default value is 0.</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_Rectangle">
      <Description>dm square or rectangle barcode: 0-square, 1-tectangle, 2-
compatible mode</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="TDCR_AppMode">
      <Description>Algorithm running mode: 0-normal mode, 1-expert mode, 2-
speed mode</Description>
      <Value>0</Value>
    </Integer>
  </Category>

  <!--Matting parameters configuration-->
  <Category Name="WayBillParam">
    <Integer Name="WAYBILL_Enable">
      <Description>Enable or not: 0-disable, 1-enable</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="WAYBILL_Ability">
      <Description>Capability sets</Description>
      <Value>7</Value>
    </Integer>
    <Integer Name="WAYBILL_Max_Width">
      <Description>The max. width supported by algorithm, range: [0,65535],
default value: 3840</Description>
      <Value>3840</Value>
```

```
    </Integer>
    <Integer Name="WAYBILL_Max_Height">
      <Description>The max. height supported by algorithm, range: [0,65535],
default value: 2748</Description>
      <Value>2748</Value>
    </Integer>
    <Integer Name="WAYBILL_OutputImageType">
      <Description>Outputted image format of waybill matting: 1-Mono8, 2-JPEG</
Description>
      <Value>2</Value>
    </Integer>
    <Integer Name="WAYBILL_JpgQuality">
      <Description>Encoding quality, range: [1,100], default value: 80</
Description>
      <Value>80</Value>
    </Integer>
    <Integer Name="WAYBILL_MinWidth">
      <Description>The min. width of waybill. Width is the longer side, and
height is the shorter side. Value range: [15,2592], the default value is 100</
Description>
      <Value>100</Value>
    </Integer>
    <Integer Name="WAYBILL_MinHeight">
      <Description>The min. height of waybill. Width is the longer side, and
height is the shorter side. Value range: [10,2048], the default value is 100</
Description>
      <Value>100</Value>
    </Integer>
    <Integer Name="WAYBILL_MaxWidth">
      <Description>The max. width of waybill, value range: [15,3072], the
default value is 3072</Description>
      <Value>3072</Value>
    </Integer>
    <Integer Name="WAYBILL_MaxHeight">
      <Description>The max. height of waybill, value range: [15,2048], the
default value is 2048</Description>
      <Value>2048</Value>
    </Integer>
    <Integer Name="WAYBILL_MorphTimes">
      <Description>Morph times, value range: [0,10], the default value is 0.</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WAYBILL_GrayLow">
      <Description>The min. gray value of barcode and characters on waybill,
value range: [0,255], the default value is 0.</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WAYBILL_GrayMid">
      <Description>The medium gray value, value range: [0,255], the default
value is 70.</Description>
      <Value>70</Value>
```

```
    </Integer>
    <Integer Name="WAYBILL_GrayHigh">
      <Description>The max. gray value of waybill background, value range:
[0,255], the default value is 130.</Description>
      <Value>130</Value>
    </Integer>
    <Integer Name="WAYBILL_BinaryAdaptive">
      <Description>Self-adaptive binaryzation, value range: [0,1], the default
value is 1.</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="WAYBILL_MaxBillBarHightRatio">
      <Description>The max. height ratio of waybill and barcode, value range:
[1,100], the default value is 20.</Description>
      <Value>20</Value>
    </Integer>
    <Integer Name="WAYBILL_MaxBillBarWidthRatio">
      <Description>The maximum width ratio of bill bar, value range: [1,100],
the default value is 5.</Description>
      <Value>5</Value>
    </Integer>
    <Integer Name="WAYBILL_MinBillBarHightRatio">
      <Description>The min. height ratio of waybill and barcode, value range:
[1,100], the default value is 5.</Description>
      <Value>5</Value>
    </Integer>
    <Integer Name="WAYBILL_MinBillBarWidthRatio">
      <Description>The min. width ratio of waybill and barcode, value range:
[1,100], the default value is 2.</Description>
      <Value>2</Value>
    </Integer>
    <Integer Name="WAYBILL_EnhanceMethod">
      <Description>Enhancement method, value range: [1,4], the default value is
2.</Description>
      <Value>2</Value>
    </Integer>
    <Integer Name="WAYBILL_ClipRatioLow">
      <Description>Enhancement low threshold ratio, value range: [0,100], the
default value is 1.</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="WAYBILL_ClipRatioHigh">
      <Description>Enhancement high threshold ratio, value range: [0,100], the
default value is 99.</Description>
      <Value>99</Value>
    </Integer>
    <Integer Name="WAYBILL_ContrastFactor">
      <Description>Contrast coefficient, value range: [1,10000], the default
value is 100.</Description>
      <Value>100</Value>
    </Integer>
    <Integer Name="WAYBILL_SharpenFactor">
```

```
      <Description>Sharpness factor, value range: [0,10000], the default value
is 0.</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WAYBILL_KernelSize">
      <Description>Size of sharpening and filter kernel, value range: [3,15],
the default value is 3.</Description>
      <Value>3</Value>
    </Integer>
    <Integer Name="WAYBILL_CodeBoundaryRow">
      <Description>Code boundary row, value range: [0,2000], the default value
is 0.</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WAYBILL_CodeBoundaryCol">
      <Description>Code boundary column, value range: [0,2000], the default
value is 0.</Description>
      <Value>0</Value>
    </Integer>
  </Category>

  <!--Weighting parameters configuration-->
  <Category Name="WeightParam">
    <Integer Name="WEIGHT_Enable">
      <Description>Enable weighing or not: 0-disable, 1-enable</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WEIGHT_DataSource">
      <Description>Set weighting data source: 1-physical serial port, 2-serial
drier</Description>
      <Value>1</Value>
    </Integer>
    <!--COM port configuration-->
    <Integer Name="WEIGHT_ComIndex">
      <Description>Serial port No., ranges from 1 to 65535</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="WEIGHT_Baudrate">
      <Description>Baudrate, ranges from 0 to 0xffffffff</Description>
      <Value>9600</Value>
    </Integer>
    <Integer Name="WEIGHT_DataBits">
      <Description>Data bits, from 5 to 8</Description>
      <Value>8</Value>
    </Integer>
    <Integer Name="WEIGHT_Parity">
      <Description>No-0, Odd-1, Even-2, Mark-3, Space-4</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WEIGHT_StopBits">
      <Description>Stop bits, values "0" or "1"</Description>
      <Value>0</Value>
```

```
    </Integer>
    <Integer Name="WEIGHT_Event">
      <Description>Whether enables weighting by event: 0-disable, 1-enable</
Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WEIGHT_DtrControl">
      <Description>Enable DTR control or not: 0-disable, 1-enable</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WEIGHT_RtsControl">
      <Description>Enable RTS control or not: 0-disable, 1-enable</Description>
      <Value>0</Value>
    </Integer>

    <!--Weight parsing precondition-->
    <Integer Name="WEIGHT_StableCount">
      <Description>Counting range when weighing, ranges form 0 to 1000, the
default value is 5.</Description>
      <Value>5</Value>
    </Integer>
    <Integer Name="WEIGHT_StablePrecision">
      <Description>Comparison error range, unit: g, ranges from 0 to 1000000</
Description>
      <Value>20</Value>
    </Integer>
    <Integer Name="WEIGHT_StableFormat">
      <Description>Enable configuring format according to precision: 0-disable,
1-enable</Description>
      <Value>0</Value>
    </Integer>
    <!--The extracted structure, which describes the process of parsing weight
value via character string-->
    <Integer Name="WEIGHT_WeighType">
      <Description>Weighing type: 0-read only, 1-echo</Description>
      <Value>0</Value>
    </Integer>
    <String Name="WEIGHT_InString">
      <Description>Inputted string in echo mode</Description>
      <Value>a</Value>
    </String>
    <Integer Name="WEIGHT_SampleRate">
      <Description>Sampling rate in echo mode, the weight reading interval time
= 1000/sampling rate</Description>
      <Value>20</Value>
    </Integer>
    <Integer Name="WEIGHT_Strlen">
      <Description>Weight data length, it values 8 by default and ranges from 1
to 32</Description>
      <Value>8</Value>
    </Integer>
    <String Name="WEIGHT_IDentifier">
```

```
      <Description>Weight data initial identifier</Description>
      <Value>=</Value>
    </String>
    <Integer Name="WEIGHT_IDentifierPos">
      <Description>The starting position of identifier in the string, ranges 
from 0 to 32</Description>
      <Value>0</Value>
    </Integer>
    <Integer Name="WEIGHT_IntegerBegin">
      <Description>The starting position of weight integer part in the string, 
ranges from 0 to 32</Description>
      <Value>4</Value>
    </Integer>
    <Integer Name="WEIGHT_IntegerEnd">
      <Description>The end position of weight integer part in the string, 
ranges from 0 to 32</Description>
      <Value>7</Value>
    </Integer>
    <Integer Name="WEIGHT_DecimalsBegin">
      <Description>The starting position of weight decimal part in the string, 
ranges from 0 to 32</Description>
      <Value>1</Value>
    </Integer>
    <Integer Name="WEIGHT_DecimalsEnd">
      <Description>The end position of weight decimal part in the string, 
ranges from 0 to 32</Description>
      <Value>2</Value>
    </Integer>
    <Integer Name="WEIGHT_Reversal">
      <Description>Whether the weighting result needs reversal: 0-no, 1-yes</
Description>
      <Value>1</Value>
    </Integer>
  </Category>

</MvLogisticsRoot>
```

A.4 Macro Definition

Maximum Value

Macro Definition	Description
MVLGS_MAX_ CODECHARATERLEN	The maximum length of code characters, the default value is 4096.
MVLGS_MAX_CODENUM	The maximum number of codes, the default value is 256.
MVLGS_MAX_CODELISTNUM	The maximum number of code lists, the default value is 24.

Exception Message

Macro Definition	Value	Description
MV_LGS_EXCEPTION_DEV_ DISCONNECT	0x00008001	The device is disconnected.
MV_LGS_EXCEPTION_ SOFTDOG_DISCONNECT	0x00008002	The softdog is disconnected.
MV_LGS_BEGIN_TRIGGER	1	Start triggering
MV_LGS_STOP_TRIGGER	0	Stop triggering

A.5 Error Code

The errors may occurred during the MvLogisticsSDK integration are listed here for reference. You can search for the detailed error descriptions according to returned error codes or error names.

Error Code

Error Code	Error Name	Description
0x00000000	MV_LGS_OK	Succeeded
0x80110000	MV_LGS_E_HANDLE	Incorrect or invalid handle
0x80110001	MV_LGS_E_SUPPORT	The function is not supported
0x80110002	MV_LGS_E_BUFOVER	Buffer is full
0x80110003	MV_LGS_E_CALLORDER	Incorrect calling sequence
0x80110004	MV_LGS_E_PARAMETER	Incorrect parameter
0x80110005	MV_LGS_E_RESOURCE	Resource request failed
0x80110006	MV_LGS_E_NODATA	No data
0x80110007	MV_LGS_E_PRECONDITION	Incorrect precondition, or running environment has changed
0x80110008	MV_LGS_E_ENCRYPT	Certificate error. No Dongle recognized, or the Dongle has expired
0x8011000a	MV_LGS_E_RULE	Filter rule error
0x80110012	MV_LGS_E_JPGENC	PG encoding error
0x80110013	MV_LGS_E_IMAGE	Image data loss, or invalid image format
0x80110014	MV_LGS_E_CONFIG	Incorrect configuration file

Error Code	Error Name	Description
0x801100FF	MV_LGS_E_UNKNOW	Unknown error
0x80112100	MV_LGS_E_CAMERA	Camera error
0x80112200	MV_LGS_E_BCR	One dimensional code error
0x80112300	MV_LGS_E_TDCR	Bar code error
0x80112400	MV_LGS_E_WAYBILL	Image matting error
0x80112500	MV_LGS_E_SCRIPT	Script rule error
0x80110100	MV_LGS_E_GC_GENERIC	Generic error
0x80110101	MV_LGS_E_GC_ARGUMENT	Invalid parameter
0x80110102	MV_LGS_E_GC_RANGE	The value is out of range
0x80110103	MV_LGS_E_GC_PROPERTY	Parameter attribute mismatched
0x80110104	MV_LGS_E_GC_RUNTIME	Running environment error
0x80110105	MV_LGS_E_GC_LOGICAL	Incorrect logic
0x80110106	MV_LGS_E_GC_ACCESS	Incorrect condition of accessing node
0x80110107	MV_LGS_E_GC_TIMEOUT	Timed out
0x80110108	MV_LGS_E_GC_ DYNAMICCAST	Cast exception
0x801101FF	MV_LGS_E_GC_UNKNOW	GenICam unknown error
0x80110200	MV_LGS_E_NOT_ IMPLEMENTED	Command is not supported
0x80110201	MV_LGS_E_INVALID_ ADDRESS	Target address does not exist
0x80110202	MV_LGS_E_WRITE_ PROTECT	Target address is not writable
0x80110203	MV_LGS_E_ACCESS_DENIED	No permission for access
0x80110204	MV_LGS_E_BUSY	Device is busy, or network is disconnected
0x80110205	MV_LGS_E_PACKET	Network packet error
0x80110206	MV_LGS_E_NETER	Network error
0x80110221	MV_LGS_E_IP_CONFLICT	IP address conflicted
0x80110300	MV_LGS_E_USB_READ	USB read error
0x80110301	MV_LGS_E_USB_WRITE	USB write error

Error Code	Error Name	Description
0x80110302	MV_LGS_E_USB_DEVICE	Device exception
0x80110303	MV_LGS_E_USB_GENICAM	GenICam error
0x80110304	MV_LGS_E_USB_ BANDWIDTH	Insufficient bandwidth
0x80110305	MV_LGS_E_USB_DRIVER	Driver is mismatched, or driver is not installed
0x801103FF	MV_LGS_E_USB_UNKNOW	USB unknown error
0x80112600	MV_LGS_E_FUSION_PARAM	Incorrect parameter(s) of fusion module
0x80112601	MV_LGS_E_FUSION_ MALLOC	Memory allocation of fusion module failed
0x80112602	MV_LGS_E_FUSION_ CALLORDER	Fusion module calling order error
0x80112603	MV_LGS_E_FUSION_ CFGFILE	Configuration file of fusion module error
0x80112604	MV_LGS_E_FUSION_ UNKNOWN	Unknown error of fusion module
0x80112605	MV_LGS_E_FUSION_ LACKBUF	Insufficient buffer of fusion module
0x80112606	MV_LGS_E_FUSION_ SUPPORT	Fusion module is not supported
0x80112700	MV_LGS_E_VOLMEASURE_ PARAM	Incorrect parameter(s) of volume measurement module
0x80112701	MV_LGS_E_VOLMEASURE_ MALLOC	Memory allocation of volume measurement module failed
0x80112702	MV_LGS_E_VOLMEASURE_ CALLORDER	Calling order of volume measurement module error
0x80112703	MV_LGS_E_VOLMEASURE_ NODATA	No data in volume measurement module
0x80112704	MV_LGS_E_VOLMEASURE_ CFGFILE	Configuration file of volume measurement module error
0x80112705	MV_LGS_E_VOLMEASURE_ NOPKG	No packet detected
0x80112706	MV_LGS_E_VOLMEASURE_ UNKNOWN	Unknown error in volume measurement module

Error Code	Error Name	Description
0x80112707	MV_LGS_E_VOLMEASURE_ LACKBUF	Insufficient buffer of volume measurement module
0x80112708	MV_LGS_E_VOLMEASURE_ SUPPORT	Volume measurement module is not supported
0x80112800	MV_LGS_E_WGHT_OPEN	Failed to start device
0x80112801	MV_LGS_E_WGHT_ENC	Weighing module encryption error
0x80112802	MV_LGS_E_WGHT_ RESOURCE	Weighing module resource initialization failed
0x80112803	MV_LGS_E_WGHT_ CALLORDER	Calling order of weighing module error
0x80112804	MV_LGS_E_WGHT_NULL	The pointer type parameter of weighing module is empty
0x80112805	MV_LGS_E_WGHT_RANGE	Incorrect value type range of weighing module
0x80112806	MV_LGS_E_WGHT_ENABLE	Incorrect capability set of weighing module
0x80112807	MV_LGS_E_WGHT_ UNKNOW	Unknown error of weighing module

A.6 Sample Code

The following sample code is for reference only.

```
#include <stdio.h>
#include <stdlib.h>

#if defined __GNUC__
#include <unistd.h>
#define Sleep(x) usleep(x##000)

#elif defined _MSC_VER
#include <windows.h>
#include <conio.h>
#endif

#include "MvLogisticsSDKDefine.h"
#include "MvLogisticsSDK.h"

//Wait for keypress
#ifdef WIN32
void WaitForKeyPress(void)
{
```

```
    while(!_kbhit())
    {
        Sleep(10);
    }
    _getch();
}
#else
void PressEnterToExit(void)
{
    int c;
    while ( (c = getchar()) != '\n' && c != EOF );
    fprintf( stderr, "\nPress enter to exit.\n");
    while( getchar() != '\n');
}
#endif

//Receive exception information in callback function
void _stdcall ExceptionCallBack(MVLGS_EXCEPTION_INFO * pstEcptInfo, void* pUser)
{
    if (NULL != pstEcptInfo)
    {
        printf("Exception ID [%d]\r\n", pstEcptInfo->nExceptionID);
        printf("Exception Device Type [%d]\r\n", pstEcptInfo->enCamType);
        printf("Exception Device SerialNum [%s]\r\n", pstEcptInfo-
>strCamSerialNum);
        printf("Exception Description [%s]\r\n", pstEcptInfo->strExceptionDes);
    }
    else
    {
        printf("Exception Info is Null!\r\n");
    }
}

//Receive package information in callback function/print package information
void _stdcall PackageCallBack(MVLGS_PACKAGE_INFO * pstPkgInfo, void* pUser)
{
    if (NULL != pstPkgInfo)
    {
        printf("/*********************************************************/\r
\n");
        if (false != pstPkgInfo->bCodeEnable)
        {
            printf("Package: Codelist Num [%d]\r\n", pstPkgInfo->nCodeListNum);

            //Display current barcode information
            for (int i = 0; i < pstPkgInfo->nCodeListNum; i++)
            {
                if (0 == pstPkgInfo->stCodeList[i].nCodeNum)
                {
                    printf("No Code Can Read!\r\n");
                }
                else
```

```
                {
                    printf("Code Num [%d]\r\n", pstPkgInfo-
>stCodeList[i].nCodeNum);
                    printf("Package: [%d]\r\n", pstPkgInfo-
>stCodeList[i].stImage.nTriggerIndex);
                    for (int j = 0; j < pstPkgInfo->stCodeList[i].nCodeNum; j++)
                    {
                        printf("Code [%d] [%s]\r\n", j, pstPkgInfo-
>stCodeList[i].stCodeInfo[j].strCode);
                    }
                }
            }
        }

        if (pstPkgInfo->bPRMEnable)
        {
            printf("Can get PRMInfo!\r\n");
        }

        //Display volume information
        if (pstPkgInfo->bVolumeEnable)
        {
            printf("Get VolumeInfo: VolumeInfo Length[%f] Width[%f] Height[%f]
Volume[%f]\r\n",
                pstPkgInfo->stVolumeInfo.fLength,
                pstPkgInfo->stVolumeInfo.fWidth,
                pstPkgInfo->stVolumeInfo.fHeight,
                pstPkgInfo->stVolumeInfo.fVolume);
        }

        //Display weight information
        if (pstPkgInfo->bWeightEnable)
        {
            printf("Get WeightInfo: WeightInfo Weight[%f]KG\r\n", pstPkgInfo-
>fWeight);
        }

        printf("/***********************************************************/\r
\n");
    }
}

int main(void)
{
    void *handle    = NULL;
    int nRet        = MV_LGS_OK;

    do
    {
        //Get MvLogistics SDK version No.
        int nVersion = MV_LGS_GetVersion();
        int nV1 = ((nVersion & 0xff000000) >> 24);
```

```
        int nV2 = ((nVersion & 0x00ff0000) >> 16);
        int nV3 = ((nVersion & 0x0000ff00) >> 8);
        int nV4 = ((nVersion & 0x000000ff));
        printf("MV LGS SDK Version: V%d.%d.%d.%d\r\n", nV1, nV2, nV3, nV4);

        //Select the device and create a handle
        nRet = MV_LGS_CreateHandle(&handle);
        if (MV_LGS_OK != nRet)
        {
            printf("Create LGS handle failed! Error Code [%#x]\r\n", nRet);
            break;
        }
        else
        {
            printf("Create LGS handle Succeed!\r\n");
        }

        //Load configuration file
        nRet = MV_LGS_LoadDevCfg(handle, "MvLogisticsSDK.xml");
        if (MV_LGS_OK != nRet)
        {
            printf("Load Device Configuration Failed! Error Code [%#x]\r\n",
nRet);
            break;
        }
        else
        {
            printf("Load Device Configuration Succeed!\r\n");
        }

        //Get package information in callback function
        nRet = MV_LGS_RegisterPackageCB(handle, PackageCallBack, handle);
        if (MV_LGS_OK != nRet)
        {
            printf("MV_LGS_RegisterPackageCB failed! Error Code [%#x]\r\n",
nRet);
            break;
        }
        else
        {
            printf("MV_LGS_RegisterPackageCB Succeed!\r\n");
        }

        //Get exception information in callback function
        nRet = MV_LGS_RegisterExceptionCB(handle, ExceptionCallBack, handle);
        if (MV_LGS_OK != nRet)
        {
            printf("MV_LGS_RegisterExceptionCB failed! Error Code [%#x]\r\n",
nRet);
            break;
        }
        else
```

```
        {
            printf("MV_LGS_RegisterExceptionCB Succeed!\r\n");
        }

        //Start stream
        nRet = MV_LGS_Start(handle);
        if (MV_LGS_OK != nRet)
        {
            printf("Start Work failed! Error Code [%#x]\r\n", nRet);
            break;
        }
        else
        {
            printf("Start Work Succeed!\r\n");
        }

        printf("Press Trigger to Get Package Information, or press a keyboard
to stop grabbing!\r\n");

#ifdef WIN32
        WaitForKeyPress();
#else
        PressEnterToExit();
#endif

        //Stop stream
        nRet = MV_LGS_Stop(handle);
        if (MV_LGS_OK != nRet)
        {
            printf("Stop Grabbing fail! Error Code [%#x]\r\n", nRet);
            break;
        }
        else
        {
            printf("Stop Grabbing Succeed!\r\n");
        }

        //Destroy the handle
        nRet = MV_LGS_DestroyHandle(handle);
        if (MV_LGS_OK != nRet)
        {
            printf("Destroy Handle fail! Error Code [%#x]\r\n", nRet);
            break;
        }
        else
        {
            if (NULL != handle)
            {
                MV_LGS_DestroyHandle(handle);
                handle = NULL;
            }
```

```
            printf("Destroy Handle Succeed!\r\n");
        }

    } while (0);

    if (NULL != handle)
    {
        MV_LGS_Stop(handle);
        MV_LGS_DestroyHandle(handle);
        handle = NULL;
    }

    printf("Press a key to exit.\r\n");

#ifdef WIN32
    WaitForKeyPress();
#else
    PressEnterToExit();
#endif

    return MV_LGS_OK;

}
```